Thiranex Internship

Customer Segmentation Project

Objective: Segment customers using demographic attributes and purchase behavior to understand distinct customer groups and support targeted marketing decisions.

1. Dataset Overview

The demonstration dataset contains 500 synthetic customer records. Features include age, gender, annual income, spending score, purchase frequency, average order value, estimated annual spend, region, preferred channel, and loyalty membership.

2. Methodology

1. Data preparation: selected demographic and behavior features and checked the dataset structure.
2. Feature scaling: standardized numerical variables so features with larger numeric ranges would not dominate clustering.
3. Clustering: applied K-Means with four clusters to group customers with similar profiles.
4. Evaluation: calculated a silhouette score as a basic measure of cluster separation.
5. Profiling: compared average age, income, spending score, and purchase frequency to create understandable segment labels.

3. Model Summary

Algorithm: K-Means clustering
Number of clusters: 4
Features: Age, Annual Income (■000s), Spending Score (1-100), Purchase Frequency (orders/year), Average Order Value (■)
Silhouette score on this synthetic dataset: 0.368

Segment	Customers	Share	Avg Income (■000s)	Avg Spending Score	Avg Orders/year
Budget / Occasional	143	28.6%	41.5	35.4	2.6
Affluent Low-Engagement	105	21.0%	72.6	34.0	3.1
High-Value Engaged	99	19.8%	85.5	73.2	7.9
High-Value Engaged	153	30.6%	42.0	72.5	9.1

4. Segment Interpretation and Actions

• High-Value Engaged: reward loyalty, provide early access, and personalize recommendations.
• Affluent Low-Engagement: use premium discovery campaigns and carefully targeted reactivation offers.
• Budget / Occasional: promote value bundles, accessible price points, and repeat-purchase incentives.
• Regular Growth: cross-sell relevant products and encourage loyalty-program enrollment.

5. Visualizations

The dashboard includes customer count by segment, income versus spending score, average purchase frequency, and average income by segment. Filters allow exploration by segment and region.

6. Deliverables

Interactive HTML dashboard, Excel analysis workbook, customer dataset CSV, segmented-customer output CSV, Python clustering script, requirements file, dashboard preview image, and this report.

Important Data Note

No real customer dataset was attached to the task screenshot. This submission pack uses synthetic records generated for demonstration and learning. Replace them with the dataset supplied by Thiranex, if available, before making claims about real customer behavior.